Submitted to the Annals of Statistics

SUPPLEMENT TO "MODEL-AGNOSTIC INTERPRETATION OF MACHINE LEARNING VIA AVERAGE MARGINAL EFFECTS"

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This supplement contains proofs of all results stated in the main paper.

References of the form Theorem 1, Lemma 2 and so on refer to the main paper. Sections and equations numbered with an S prefix are internal to this supplement.

S1. Proof of Lemma 1 (Riesz Representation).

PROOF. Recall $q = wp$ extended by zero outside Ω, so that q is defined on all of $\mathbb{R}^d$, $\|q\|_\infty \le p_{\max}$, and $q(x) \neq 0$ implies $x \in \operatorname{supp}(w) \subseteq \Omega_{j,h}$. Fix $g \in L^1(P)$ and consider the positive-shift half of the finite difference:

$$\mathbb{E}\left[w(X)\, g(X + he_j)\right] = \int_{\mathbb{R}^d} q(x)\, g(x + he_j)\, dx = \int_{\mathbb{R}^d} q(u - he_j)\, g(u)\, du \tag{S1.1}$$

by the substitution $u = x + he_j$. The manipulation is legitimate because on the set where the integrand is nonzero we have $x \in \Omega_{j,h}$, hence $u = x + he_j \in \Omega$, so g is evaluated only on Ω; and the integral is absolutely convergent since:

$$\int q(u - he_j)\ |g(u)|\ du \le p_{\max} \int_\Omega |g(u)|\ du \le \frac{p_{\max}}{p_{\min}}\, \mathbb{E}\,|g(X)| < \infty \tag{S1.2}$$

The negative-shift half is identical with $u = x - he_j$. Subtracting and dividing by $2h$:

$$\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] = \int_\Omega \frac{q(u - he_j) - q(u + he_j)}{2h}\, g(u)\, du = \mathbb{E}\left[\alpha_h(X)\, g(X)\right] \tag{S1.3}$$

where the restriction of the domain of integration to Ω in the second step uses that g is supported there, and the final step multiplies and divides by $p(u) \ge p_{\min} > 0$. The uniform bound follows from $|q(u - he_j) - q(u + he_j)| \le 2p_{\max}$ and $p(u) \ge p_{\min}$. Applying the representation to $g \equiv 1$ gives $\mathbb{E}[\alpha_h(X)] = \mathbb{E}[w(X)\Delta_{j,h}1] = 0$ since the finite difference of a constant vanishes. □

S2. Proof of Proposition 1 (Recovery of the Average Derivative).

PROOF. Let $f \in W^{1,1}(\Omega)$ and fix $h \le h_0$, so that $\operatorname{supp}(w) \subseteq \Omega_{j,h_0} \subseteq \Omega_{j,h}$ and every segment $\{x + te_j : |t| \le h\}$ with $x \in \operatorname{supp}(w)$ lies in Ω. Sobolev functions are absolutely continuous on almost every line parallel to a coordinate axis, with the classical derivative along the line agreeing almost everywhere with the weak derivative (Evans and Gariepy, 2015, Section 4.9). Hence for almost every $x \in \operatorname{supp}(w)$:

$$f(x + he_j) - f(x - he_j) = \int_{-h}^{h} \partial_j f(x + te_j)\, dt \tag{S2.1}$$

Keywords and phrases. average marginal effects, Neyman orthogonality, semiparametric efficiency.

and therefore:

$$\theta_h(f) - \mathbb{E}\left[w(X)\,\partial_j f(X)\right] = \int_{\mathbb{R}^d} q(x)\,\frac{1}{2h}\int_{-h}^{h}\left[\partial_j f(x+te_j) - \partial_j f(x)\right]dt\,dx \tag{S2.2}$$

Bounding q by $p_{\max}$, applying Tonelli, and extending $\partial_j f$ by zero outside Ω:

$$|\theta_h(f) - \mathbb{E}\left[w\,\partial_j f\right]| \le p_{\max}\sup_{|t|\le h}\int_{\mathbb{R}^d}|\partial_j f(x+te_j) - \partial_j f(x)|\;\mathbf{1}\{x\in\mathrm{supp}(w)\}\,dx \tag{S2.3}$$

where the extension introduces no error because $x + te_j \in \Omega$ for $x \in \mathrm{supp}(w)$ and $|t| \le h \le h_0$. Continuity of translation in $L^1(\mathbb{R}^d)$ applied to the extension of $\partial_j f$ sends the right side to zero as $h \downarrow 0$. For the second claim, $|w_m \partial_j f| \le |\partial_j f|$, which is P-integrable because $\partial_j f \in L^1(\Omega)$ and $p \le p_{\max}$; dominated convergence gives $\mathbb{E}[w_m \partial_j f] \to \mathbb{E}[\partial_j f] = \mathrm{AME}_j$. □

S3. Proof of Lemma 2 (Exact Product Bias).

PROOF. Conditioning on X and using $\mathbb{E}[Y \mid X] = f(X)$:

$$\mathbb{E}\left[\tilde{\alpha}(X)\,(Y - \tilde{f}(X))\right] = \mathbb{E}\left[\tilde{\alpha}(X)\,(f(X) - \tilde{f}(X))\right] \tag{S3.1}$$

By linearity of $\Delta_{j,h}$ and Lemma 1 applied to $g = \tilde{f} - f \in L^1(P)$:

$$\mathbb{E}\left[w(X)\,\Delta_{j,h}\tilde{f}(X)\right] - \theta_h(f) = \mathbb{E}\left[w(X)\,\Delta_{j,h}(\tilde{f} - f)(X)\right] = \mathbb{E}\left[\alpha_h(X)\,(\tilde{f}(X) - f(X))\right] \tag{S3.2}$$

Summing the two displays gives:

$$\mathbb{E}\left[\psi(X, Y; \tilde{f}, \tilde{\alpha}, \theta_h(f))\right] = \mathbb{E}\left[(\alpha_h(X) - \tilde{\alpha}(X))(\tilde{f}(X) - f(X))\right] \tag{S3.3}$$

and the bound is Cauchy–Schwarz. □

S4. Proof of Theorem 1 (Asymptotic Linearity and Normality).

S4.1. *Step 0: The Trimmed Difference Operator is Bounded on $L^2(P)$.* For any $g \in L^2(\Omega)$, using $(a-b)^2 \le 2a^2 + 2b^2$ and the change of variables from the proof of Lemma 1:

$$\mathbb{E}\left[\left(w(X)\,\Delta_{j,h}g(X)\right)^2\right] \le \frac{1}{2h^2}\left(\mathbb{E}\left[w(X)^2 g(X + he_j)^2\right] + \mathbb{E}\left[w(X)^2 g(X - he_j)^2\right]\right)$$

$$\le \frac{1}{h^2}\cdot\frac{p_{\max}}{p_{\min}}\,\mathbb{E}\left[g(X)^2\right] \tag{S4.1}$$

which establishes (3.19) with constant $\kappa = h^{-1}(p_{\max}/p_{\min})^{1/2}$. This is the only property of the finite difference used below; no derivative of g appears anywhere.

S4.2. *Step 1: Decomposition by Folds.* Write $\theta = \theta_h(f)$, $\psi_i = \psi(\mathbf{x}_i, y_i; f, \alpha_h, \theta)$, and for $i \in I_k$ let $\hat{\psi}_i = \psi(\mathbf{x}_i, y_i; \hat{f}^{(-k)}, \hat{\alpha}^{(-k)}, \theta)$. Then:

$$\sqrt{n}\,(\hat{\theta}_h - \theta) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i + \sum_{k=1}^{K}\sqrt{\frac{n_k}{n}}\left(\sqrt{n_k}\,A_k + \sqrt{n_k}\,B_k\right) \tag{S4.2}$$

where, with $\mathbb{E}_k$ denoting expectation over a fresh draw (X, Y) holding the fold-k nuisances fixed:

$$A_k = \mathbb{E}_k\left[\hat{\psi} - \psi\right], \qquad B_k = \frac{1}{n_k}\sum_{i\in I_k}\left(\hat{\psi}_i - \psi_i - A_k\right) \tag{S4.3}$$

Since K is fixed and $n_k/n \to 1/K$, it suffices that $\sqrt{n_k}A_k = o_p(1)$ and $\sqrt{n_k}B_k = o_p(1)$ for each k.

S4.3. *Step 2: The Bias Term.* By Lemma 2 with $(\tilde{f}, \tilde{\alpha}) = (\hat{f}^{(-k)}, \hat{\alpha}^{(-k)})$:

$$|\sqrt{n_k}\, A_k| \le \sqrt{n_k}\, \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2} \cdot \left\|\hat{f}^{(-k)} - f\right\|_{P,2} = o_p(1) \tag{S4.4}$$

by Assumption 3(b). This is the only place the product rate is used, and it is used without any expansion: the bound is exact.

S4.4. *Step 3: The Empirical Process Term.* Conditional on the out-of-fold data, the summands of B_k are i.i.d. and mean zero, so:

$$\mathbb{E}\left[(\sqrt{n_k}\, B_k)^2 \mid \text{out-of-fold}\right] \le \mathbb{E}_k\left[(\hat{\psi} - \psi)^2\right] \tag{S4.5}$$

Decompose, adding and subtracting $\hat{\alpha}^{(-k)}(x)(y - f(x))$:

$$\hat{\psi} - \psi = w\,\Delta_{j,h}(\hat{f}^{(-k)} - f) + (\hat{\alpha}^{(-k)} - \alpha_h)(y - f) - \hat{\alpha}^{(-k)}\,(\hat{f}^{(-k)} - f) \tag{S4.6}$$

and bound each piece in $L^2(P)$. The middle piece is bounded by $\bar{\sigma}\, \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}$ after conditioning on X under either assumption set, since it uses only the conditional variance bound of Assumption 1. The first and third pieces are where the two assumption sets diverge.

Under the density bounds of Assumption 1, the first is bounded by $\kappa \left\|\hat{f}^{(-k)} - f\right\|_{P,2}$ via (S4.1) and the third by $\bar{A} \left\|\hat{f}^{(-k)} - f\right\|_{P,2}$ using Assumption 3(c). All three are then $o_p(1)$ by Assumption 3(a).

Under Assumption 2 neither bound is available: the representer is unbounded, and the finite-difference operator is genuinely unbounded on $L^2(P)$ rather than merely unbounded by this argument. For independent standard normal covariates, taking $g_a(x) = e^{ax_j}$ gives $\|g_a\|_{P,2}^2 = e^{2a^2}$ and $\|g_a(\cdot + he_j)\|_{P,2}^2 = e^{2ah}e^{2a^2}$, so the operator norm is $\sup_a e^{ah} = \infty$. Both pieces are instead bounded by Hölder. Writing $\Delta = \hat{f}^{(-k)} - f$ and letting r be as in (3.2) with conjugate r', a change of variables gives

$$\mathbb{E}\left[\Delta(X \pm he_j)^2\right] = \mathbb{E}\left[\Delta(X)^2\, \frac{p(X \mp he_j)}{p(X)}\right] \le \|\Delta\|_{P,2r'}^2 \left(\mathbb{E}\left[\left(\frac{p(X \mp he_j)}{p(X)}\right)^r\right]\right)^{1/r} \tag{S4.7}$$

which is $o_p(1)$ by Assumption 3(a) in its $L^{2+\delta}$ form, with $2r' = 2 + \delta$; the first piece follows from (S4.7) and $|w| \le 1$. For the third, $\mathbb{E}[(\hat{\alpha}^{(-k)})^2\Delta^2] \le \left\|\hat{\alpha}^{(-k)}\right\|_{P,q}^2 \|\Delta\|_{P,2s'}^2$ by Hölder with $s = q/2$, which is $o_p(1)$ by Assumption 3(c) in its moment form together with (a).

Either way $\mathbb{E}_k[(\hat{\psi} - \psi)^2] = o_p(1)$, and conditional Chebyshev gives $\sqrt{n_k}\, B_k = o_p(1)$. Note that only consistency of the nuisances is used here, never a rate; the rate enters at Step 2 alone.

S4.5. *Step 4: Central Limit Theorem and Variance Estimation.* The ψ_i are i.i.d., mean zero by Lemma 2 at $(\tilde{f}, \tilde{\alpha}) = (f, \alpha_h)$, with variance:

$$V_h = \mathbb{E}[\psi_i^2] \le 2\,\mathbb{E}\left[(w\Delta_{j,h}f)^2\right] + 2\,\|\alpha_h\|_\infty^2\, \mathbb{E}[\sigma^2(X)] < \infty \tag{S4.8}$$

by (S4.1) and Lemma 1 under the density bounds of Assumption 1. Under Assumption 2 the same finiteness follows from (S4.7) applied to f, which is bounded, together with $\alpha_h \in L^2(P)$ from Assumption 2(c) and $\mathbb{E}[Y^2] < \infty$; for the Gaussian design the representer's second moment is (3.3). Either way $\mathrm{Var}(\psi) < \infty$ and the classical CLT applies. For the variance estimator, write $\hat{\psi}_i(\hat{\theta}_h) = \hat{\psi}_i - (\hat{\theta}_h - \theta)$; then:

$$\hat{V}_h = \frac{1}{n}\sum_i \hat{\psi}_i^2 - 2(\hat{\theta}_h - \theta)\,\frac{1}{n}\sum_i \hat{\psi}_i + (\hat{\theta}_h - \theta)^2 \tag{S4.9}$$

The last two terms vanish in probability since $\hat{\theta}_h - \theta = o_p(1)$ and $n^{-1}\sum_i \hat{\psi}_i = O_p(n^{-1/2})$ by Steps 1–3. For the first term:

$$\left|\frac{1}{n}\sum_i \hat{\psi}_i^2 - \frac{1}{n}\sum_i \psi_i^2\right| \le \frac{1}{n}\sum_i (\hat{\psi}_i - \psi_i)^2 + 2\left(\frac{1}{n}\sum_i (\hat{\psi}_i - \psi_i)^2\right)^{1/2}\left(\frac{1}{n}\sum_i \psi_i^2\right)^{1/2} \tag{S4.10}$$

by Cauchy–Schwarz; $n^{-1}\sum_i(\hat{\psi}_i - \psi_i)^2 = o_p(1)$ by the fold-wise conditional Markov inequality and the Step 3 bound, and $n^{-1}\sum_i \psi_i^2 \xrightarrow{p} V_h$ by the law of large numbers. Hence $\hat{V}_h \xrightarrow{p} V_h$. □

S5. Proof of Proposition 2 (Double Robustness). The argument retraces the proof of Theorem 1 with the pseudo-true influence function in place of the efficient one; we record the steps that change.

S5.1. *Step 1: The Pseudo-True Influence Function.* Define $\psi_{g,i} = w(\mathbf{x}_i)\,\Delta_{j,h}g(\mathbf{x}_i) + \alpha_h(\mathbf{x}_i)\,(y_i - g(\mathbf{x}_i)) - \theta_h(f)$. By Lemma 2 with $(\tilde{f}, \tilde{\alpha}) = (g, \alpha_h)$:

$$\mathbb{E}[\psi_{g,i}] = \mathbb{E}\big[(\alpha_h - \alpha_h)(g - f)\big] = 0 \tag{S5.1}$$

for every $g \in L^2(\Omega)$, which also establishes $\mathbb{E}[m_g(X)] = \theta_h(f)$ after conditioning on X. Conditioning likewise gives $\mathbb{E}[\psi_{g,i} \mid X] = m_g(X) - \theta_h(f)$ and $\mathrm{Var}(\psi_{g,i} \mid X) = \alpha_h(X)^2\sigma^2(X)$, so $\mathrm{Var}(\psi_{g,i}) = V_h(g)$ by the law of total variance; finiteness follows from (S4.1), $g \in L^2(\Omega)$, the bound on α_h from Lemma 1, and $\mathbb{E}[Y^2] < \infty$ under the density bounds; under Assumption 2 it follows from (S4.7), $\alpha_h \in L^2(P)$, and $\mathbb{E}[Y^2] < \infty$.

S5.2. *Step 2: The Bias Term.* With $A_k = \mathbb{E}_k[\hat{\psi} - \psi_g] = \mathbb{E}_k[\hat{\psi}]$, Lemma 2 gives $A_k = \mathbb{E}[(\alpha_h - \hat{\alpha}^{(-k)})(\hat{f}^{(-k)} - f)]$, so:

$$\begin{aligned}|\sqrt{n_k}\,A_k| &\le \sqrt{n_k}\,\left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}\left(\left\|\hat{f}^{(-k)} - g\right\|_{P,2} + \|g - f\|_{P,2}\right) \\ &= \sqrt{n_k}\,o_p(n^{-1/2})\cdot O_p(1) = o_p(1)\end{aligned} \tag{S5.2}$$

by conditions (a) and (b); the second factor is $O_p(1)$ rather than $o_p(1)$ because g need not equal f, and it is exactly here that condition (b) must be strengthened to $o_p(n^{-1/2})$ relative to Assumption 3(b). When $\hat{\alpha}^{(-k)} = \alpha_h$, $A_k = 0$ identically at every n; averaging the fold-wise identity $\mathbb{E}[\hat{\psi}_i + \theta_h(f) \mid \text{out-of-fold data}] = \theta_h(f)$, which holds because the observations in fold k are independent of $\hat{f}^{(-k)}$, gives the exact unbiasedness claim.

S5.3. *Step 3: The Empirical Process Term.* Decompose, for $i \in I_k$:

$$\hat{\psi}_i - \psi_{g,i} = w\,\Delta_{j,h}(\hat{f}^{(-k)} - g) + (\hat{\alpha}^{(-k)} - \alpha_h)(y_i - \hat{f}^{(-k)}) + \alpha_h\,(g - \hat{f}^{(-k)}) \tag{S5.3}$$

The first and third terms have conditional $L^2(P)$ norms bounded by $\kappa\left\|\hat{f}^{(-k)} - g\right\|_{P,2}$ and $\|\alpha_h\|_\infty\left\|\hat{f}^{(-k)} - g\right\|_{P,2}$ respectively, both $o_p(1)$ by condition (a). For the middle term, write $y - \hat{f}^{(-k)} = (y - f) + (f - g) + (g - \hat{f}^{(-k)})$ and bound the three resulting pieces: the first by $\bar{\sigma}^2\left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}^2 = o_p(1)$ after conditioning on X; the third by $(\bar{A} +$

$\|\alpha_h\|_\infty)^2 \left\|\hat{f}^{(-k)} - g\right\|_{P,2}^2 = o_p(1)$; and the second piece, $\mathbb{E}[(\hat{\alpha}^{(-k)} - \alpha_h)^2 (f-g)^2]$, tends to zero. Under the density bounds this is dominated convergence along subsequences: the factor $(\hat{\alpha}^{(-k)} - \alpha_h)^2$ is bounded by $(\bar{A} + \|\alpha_h\|_\infty)^2$ and converges to zero in P-measure by condition (b), while $(f-g)^2 \in L^1(P)$ dominates. Under Assumption 2 the domination is unavailable and Hölder is used instead: $\mathbb{E}[(\hat{\alpha}^{(-k)} - \alpha_h)^2 (f-g)^2] \le \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,q}^2 \|f-g\|_{P,2t'}^2$ with $t = q/2$, which requires the fixed approximation error $f-g$ to lie in $L^{2t'}(P)$ — a mild condition, and one that holds whenever g and f are bounded. Conditional Chebyshev then gives $\sqrt{n_k}\, B_k = o_p(1)$ exactly as in Step 3 of the proof of Theorem 1.

S5.4. *Step 4: Conclusion.* The $\psi_{g,i}$ are i.i.d., mean zero, with finite positive variance $V_h(g)$, so the classical CLT applies to their normalized sum, and Steps 2–3 identify $\sqrt{n}(\hat{\theta}_h - \theta_h(f))$ with that sum up to $o_p(1)$. The consistency of $\hat{V}_h$ for $V_h(g)$ repeats Step 4 of the proof of Theorem 1 verbatim with ψ_g in place of ψ, using the Step 3 bound just established. Finally, when $g \neq f$ the influence function ψ_g differs from the efficient influence function ψ of Theorem 2; since in the nonparametric model every regular asymptotically linear estimator has influence function equal to the efficient one, the estimator is not regular in this case, and no conflict with the convolution theorem arises. □

S6. Proof of Theorem 2 (Semiparametric Efficiency).

PROOF. Consider one-dimensional submodels $dP_t = (1 + ts)\, dP$ indexed by bounded functions $s \in L_0^2(P) = \{s : \mathbb{E}[s(X,Y)] = 0\}$, which are well defined for $|t| < 1/\|s\|_\infty$ and differentiable in quadratic mean at $t = 0$ with score s; bounded mean-zero functions are dense in $L_0^2(P)$, so the tangent space of the nonparametric model is all of $L_0^2(P)$ (van der Vaart, 1998, Chapter 25). Write $\bar{s}_X(x) = \mathbb{E}[s(X,Y) \mid X = x]$ for the induced score of the covariate marginal and $f_t(x) = \mathbb{E}_t[Y \mid X = x]$ for the regression function along the path. Differentiating the Bayes formula for f_t at $t = 0$ gives, for P-almost every x:

$$\dot{f}(x) := \frac{\partial}{\partial t} f_t(x)\Big|_{t=0} = \mathbb{E}\left[(Y - f(X))\, s(X,Y) \mid X = x\right] \tag{S6.1}$$

where differentiation under the integral is justified by $\mathbb{E}[Y^2] < \infty$ and the boundedness of s. The functional along the path is $\theta(t) = \mathbb{E}_t[w(X)\, \Delta_{j,h} f_t(X)]$, and the product rule gives:

$$\frac{d\theta}{dt}\Big|_{t=0} = \mathbb{E}\left[w\, \Delta_{j,h} f(X) \cdot \bar{s}_X(X)\right] + \mathbb{E}\left[w\, \Delta_{j,h} \dot{f}(X)\right] \tag{S6.2}$$

For the first term, note $\mathbb{E}[\bar{s}_X] = 0$, so the term equals $\mathbb{E}[(w\Delta_{j,h} f(X) - \theta_h(f))\, \bar{s}_X(X)] = \mathbb{E}[(w\Delta_{j,h} f(X) - \theta_h(f))\, s(X,Y)]$, the last equality because $w\Delta_{j,h} f(X) - \theta_h(f)$ is X-measurable. For the second term, $\dot{f} \in L^1(P)$ (indeed L^2) since s is bounded, so Lemma 1 applies:

$$\mathbb{E}\left[w\, \Delta_{j,h} \dot{f}(X)\right] = \mathbb{E}\left[\alpha_h(X)\, \dot{f}(X)\right] = \mathbb{E}\left[\alpha_h(X)\, (Y - f(X))\, s(X,Y)\right] \tag{S6.3}$$

by the tower property. Summing:

$$\frac{d\theta}{dt}\Big|_{t=0} = \mathbb{E}\left[\psi(X,Y; f, \alpha_h, \theta_h(f))\, s(X,Y)\right] \tag{S6.4}$$

for every score s, so θ_h is pathwise differentiable with influence function ψ; since $\psi \in L_0^2(P)$ lies in the tangent space, it is the efficient influence function and the bound is $\mathbb{E}[\psi^2]$ (van der Vaart, 1998, Theorem 25.20). The cross term in $\mathbb{E}[\psi^2]$ vanishes by conditioning on X, yielding (3.22). Attainment follows from Theorem 1, whose expansion exhibits $\hat{\theta}_h$ as asymptotically linear with influence function ψ, hence regular and efficient (van der Vaart, 1998, Lemma 25.23). □

S7. Proof of Lemma 3 (Derivative Existence, Sobolev).

S7.1. *What Needs to Be Shown.* That $\hat{f}_n \in W^{2,p}(\Omega)$ for $p > d$ implies $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$ with the norm bound:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S7.1}$$

REMARK 1. The condition $p > d$ is required by the proof. It is the condition under which Theorem 6.5 of Evans and Gariepy (2015) guarantees pointwise differentiability of $W^{1,p}_{loc}$ functions. This is slightly stronger than the condition $p > 1$ stated in the main text and is an explicit assumption of the lemma.

S7.2. *Step 1: Existence of the Gradient as an L^p Function.* Since $\hat{f}_n \in W^{2,p}(\Omega)$ and $p > d$, we have in particular that $\hat{f}_n \in W^{1,p}(\Omega)$. Extending $\hat{f}_n$ to a $W^{1,p}_{loc}(\mathbb{R}^d)$ function by a standard extension operator (Evans and Gariepy, 2015, Section 4.4), we may apply Theorem 6.5 of Evans and Gariepy (2015):

> *Theorem 6.5 (Evans and Gariepy, 2015).* Assume $f \in W^{1,p}_{loc}(\mathbb{R}^n)$ for some $n < p \leq \infty$. Then f is differentiable $\mathcal{L}^n$-a.e., and its classical derivative equals its weak derivative $\mathcal{L}^n$-a.e.

Applying this theorem to $\hat{f}_n$ with $n = d$ and $p > d$, we conclude that $\hat{f}_n$ is classically differentiable at $\mathcal{L}^d$-almost every $x \in \Omega$, and:

$$\nabla \hat{f}_n(x) = D\hat{f}_n(x) \quad \mathcal{L}^d\text{-a.e.} \tag{S7.2}$$

where $D\hat{f}_n$ denotes the weak gradient of $\hat{f}_n$. Since $\hat{f}_n \in W^{1,p}(\Omega)$, the weak gradient $D\hat{f}_n \in L^p(\Omega)^d$ by definition of the Sobolev space. Therefore $\nabla \hat{f}_n \in L^p(\Omega)^d$.

Applying the same argument to each first-order weak derivative $\partial_j \hat{f}_n \in W^{1,p}(\Omega)$ — which belongs to $W^{1,p}(\Omega)$ because $\hat{f}_n \in W^{2,p}(\Omega)$ — we conclude that each $\partial_j \hat{f}_n$ is classically differentiable $\mathcal{L}^d$-a.e. with classical second derivatives in $L^p(\Omega)$. Therefore $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$.

S7.3. *Step 2: The Norm Bound.* We establish the chain of inequalities:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S7.3}$$

First Inequality. By definition of the $W^{1,p}(\Omega)$ norm:

$$\left\|\hat{f}_n\right\|_{W^{1,p}} = \left\|\hat{f}_n\right\|_{L^p} + \sum_{j=1}^{d} \left\|\partial_j \hat{f}_n\right\|_{L^p} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} \tag{S7.4}$$

Since $\left\|\hat{f}_n\right\|_{L^p} \geq 0$:

$$\left\|\hat{f}_n\right\|_{W^{1,p}} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} \geq \left\|\nabla \hat{f}_n\right\|_{L^p} \tag{S7.5}$$

Second Inequality. By definition of the $W^{2,p}(\Omega)$ norm using multi-index notation:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} \tag{S7.6}$$

The multi-indices with $|\alpha| \leq 1$ form a strict subset of those with $|\alpha| \leq 2$, and all terms are nonnegative, so:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} \geq \sum_{|\alpha| \leq 1} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} = \left\|\hat{f}_n\right\|_{W^{1,p}} \tag{S7.7}$$

S7.4. *Conclusion.* For $p > d$, membership of $\hat{f}_n$ in $W^{2,p}(\Omega)$ implies via Theorem 6.5 of Evans and Gariepy (2015) that $\nabla \hat{f}_n$ exists $\mathcal{L}^d$-almost everywhere as a classical gradient, coincides with the weak gradient, and belongs to $W^{1,p}(\Omega)^d \subset L^p(\Omega)^d$. The gradient operator $\nabla : W^{2,p}(\Omega) \to W^{1,p}(\Omega)^d$ is bounded with:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S7.8}$$

□

S8. Proof of Lemma 4 (Derivative Consistency, Sobolev).

S8.1. *What Needs to Be Shown.* That well-behavedness in the Sobolev sense implies $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

S8.2. *Proof Strategy.* The proof uses a subsequence argument. We show that every subsequence of $\{\hat{f}_n\}$ has a further subsequence along which $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \to 0$ almost surely. By the standard metric space characterization of convergence in probability — a sequence converges in probability if and only if every subsequence has an almost surely convergent further subsequence — this implies $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

Suppose for contradiction that $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p}$ does not converge to zero in probability. Then there exist $\epsilon > 0$ and $\delta > 0$ and a subsequence $\{n_k\}$ such that:

$$\mathbb{P}\left(\left\|\nabla \hat{f}_{n_k} - \nabla f\right\|_{L^p} > \epsilon\right) > \delta \quad \forall k \tag{S8.1}$$

We derive a contradiction by extracting a further subsequence along which the gradient converges to zero almost surely.

S8.3. *Step 1: Extract a Strongly Convergent Subsequence.* Take any subsequence $\{\hat{f}_{n_k}\}$. By the uniform $W^{2,p}(\Omega)$ boundedness condition:

$$\sup_k \left\|\hat{f}_{n_k}\right\|_{W^{2,p}} < \infty \quad \text{almost surely} \tag{S8.2}$$

The Rellich–Kondrachov compact embedding theorem states that for bounded Lipschitz Ω, the embedding $W^{2,p}(\Omega) \hookrightarrow\hookrightarrow W^{1,p}(\Omega)$ is compact (Evans, 2010, Theorem 5, Section 5.7). Compactness of the embedding means that every sequence bounded in $W^{2,p}(\Omega)$ has a subsequence converging strongly in $W^{1,p}(\Omega)$. Therefore there exists a further subsequence $\{\hat{f}_{n_{k_j}}\}$ and a function $g \in W^{1,p}(\Omega)$ such that:

$$\left\|\hat{f}_{n_{k_j}} - g\right\|_{W^{1,p}} \to 0 \quad \text{almost surely as } j \to \infty \tag{S8.3}$$

S8.4. *Step 2: Identify the Limit.* Strong $W^{1,p}(\Omega)$ convergence implies strong L^p convergence, since by definition of the $W^{1,p}$ norm:

$$\left\|\hat{f}_{n_{k_j}} - g\right\|_{L^p} \leq \left\|\hat{f}_{n_{k_j}} - g\right\|_{W^{1,p}} \to 0 \quad \text{almost surely} \tag{S8.4}$$

Therefore $\hat{f}_{n_{k_j}} \to g$ in L^p almost surely. By the L^p consistency assumption, $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$, so in particular along the subsequence $\hat{f}_{n_{k_j}} \to f$ in L^p in probability, and hence almost surely along a further subsequence if necessary. By uniqueness of limits in $L^p(\Omega)$:

$$g = f \quad \mathcal{L}^d\text{-a.e. on } \Omega \tag{S8.5}$$

Therefore the strongly convergent subsequence satisfies:

$$\left\|\hat{f}_{n_{k_j}} - f\right\|_{W^{1,p}} \to 0 \quad \text{almost surely} \tag{S8.6}$$

S8.5. *Step 3: Extract Gradient Convergence.* Since $\hat{f}_{n_{k_j}} \to f$ strongly in $W^{1,p}(\Omega)$, expanding the $W^{1,p}$ norm:

$$\left\|\hat{f}_{n_{k_j}} - f\right\|_{W^{1,p}} = \left\|\hat{f}_{n_{k_j}} - f\right\|_{L^p} + \left\|\nabla\hat{f}_{n_{k_j}} - \nabla f\right\|_{L^p} \to 0 \quad \text{almost surely} \tag{S8.7}$$

Since both terms are nonnegative and their sum goes to zero, each term goes to zero individually. In particular:

$$\left\|\nabla\hat{f}_{n_{k_j}} - \nabla f\right\|_{L^p} \to 0 \quad \text{almost surely as } j \to \infty \tag{S8.8}$$

S8.6. *Conclusion.* We have shown that from any subsequence $\{\hat{f}_{n_k}\}$ we can extract a further subsequence $\{\hat{f}_{n_{k_j}}\}$ along which $\left\|\nabla\hat{f}_{n_{k_j}} - \nabla f\right\|_{L^p} \to 0$ almost surely. This contradicts the assumption that $\mathbb{P}(\left\|\nabla\hat{f}_{n_k} - \nabla f\right\|_{L^p} > \epsilon) > \delta$ for all k. Therefore the assumption was false and:

$$\left\|\nabla\hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{S8.9}$$

which is the conclusion of Lemma 4. □

REMARK 2. The key step is the application of Rellich–Kondrachov in Step 1. The theorem requires Ω to be bounded and Lipschitz — both of which are assumed throughout — and $p > 1$, which is assumed in the lemma statement. The compact embedding $W^{2,p} \hookrightarrow\hookrightarrow W^{1,p}$ is strictly stronger than the continuous embedding $W^{2,p} \hookrightarrow W^{1,p}$: the latter follows from the norm bound in Lemma 3, while the former provides the compactness needed to extract convergent subsequences. It is this compactness, not merely the norm bound, that allows the interchange of limit and gradient in the proof.

S9. Proof of Lemma 5 (Derivative Existence, BV).

S9.1. *What Needs to Be Shown.* Two things: that BV membership implies $D\hat{f}_n$ exists as a Radon measure with finite total variation, and the explicit formula for piecewise constant functions.

S9.2. *Step 1: Existence of $D\hat{f}_n$ as a Radon Measure.* We establish existence in three steps, using the Riesz representation theorem and the Radon-Nikodym theorem.

Substep 1a: The Distributional Gradient as a Linear Functional. For any $\hat{f}_n \in BV(\Omega)$, the distributional gradient is defined as the linear functional $\Lambda : C_c^1(\Omega)^d \to \mathbb{R}$ given by:

$$\Lambda(\phi) = -\int_\Omega \hat{f}_n \operatorname{div}\phi \, dx \quad \forall \phi \in C_c^1(\Omega)^d \tag{S9.1}$$

By the definition of $BV(\Omega)$, this functional has finite total variation:

$$|\Lambda|(\Omega) = \sup\left\{\Lambda(\phi) : \phi \in C_c^1(\Omega)^d,\ \|\phi\|_{L^\infty} \le 1\right\} < \infty \tag{S9.2}$$

This is precisely the condition that $\hat{f}_n \in BV(\Omega)$ imposes — the distributional gradient is a bounded linear functional on $C_c^1(\Omega)^d$ with respect to the supremum norm.

Substep 1b: Riesz Representation. Since Λ is a bounded linear functional on $C_c^1(\Omega)^d$ with finite total variation, the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) guarantees the existence of a vector-valued Radon measure $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ such that:

$$\Lambda(\phi) = \int_\Omega \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \tag{S9.3}$$

That is:

$$-\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx = \int_\Omega \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \tag{S9.4}$$

This is the distributional gradient equation, and $D\hat{f}_n$ is the distributional gradient of $\hat{f}_n$ represented as a Radon measure.

Substep 1c: Finiteness of Total Variation. By the Radon-Nikodym theorem for Radon measures (Evans and Gariepy, 2015, Theorem 1.29), the derivative $D_{\mathcal{L}^d}(D\hat{f}_n)$ exists and is finite $\mathcal{L}^d$-almost everywhere on Ω. Since Ω is bounded and the total variation of Λ is finite by the BV assumption, we conclude:

$$|D\hat{f}_n|(\Omega) = |\Lambda|(\Omega) < \infty \tag{S9.5}$$

Therefore $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ with $|D\hat{f}_n|(\Omega) < \infty$, as claimed.

S9.3. *Step 2: Explicit Formula for Piecewise Constant Functions.* Let $\hat{f}_n = \sum_j \bar{y}_j \mathbf{1}_{R_j}$ where $\{R_j\}$ is a finite partition of Ω into sets with Lipschitz boundaries, and $\bar{y}_j$ denotes the response mean on leaf R_j. We derive the explicit formula for $D\hat{f}_n$ by applying the Gauss–Green theorem to each leaf.

Substep 2a: Applying the Definition. For any test function $\phi \in C_c^1(\Omega)^d$, the distributional gradient equation from Substep 1b gives:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = -\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx = -\sum_j \bar{y}_j \int_{R_j} \operatorname{div} \phi \, dx \tag{S9.6}$$

where the second equality uses the piecewise constant structure of $\hat{f}_n$ and linearity of the integral.

Substep 2b: Gauss–Green on Each Leaf. Since each R_j has Lipschitz boundary, the Gauss–Green theorem for BV functions (Evans and Gariepy, 2015, Theorem 5.16) applies to each leaf:

$$\int_{R_j} \operatorname{div} \phi \, dx = \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \tag{S9.7}$$

where ν_j denotes the outward unit normal to ∂R_j and $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional Hausdorff measure on the boundary.

Substep 2c: Cancellation on Interior Boundaries. Substituting the Gauss–Green formula into the expression from Substep 2a:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = -\sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \tag{S9.8}$$

Every interior boundary $\partial R_j \cap \partial R_k$ between adjacent leaves j and k appears twice in this sum — once with outward normal ν_j for leaf R_j and once with outward normal $\nu_k = -\nu_j$ for leaf R_k. Collecting these paired contributions and writing $\nu_{jk} = \nu_j$ for the unit normal pointing from R_j into R_k:

$$\begin{aligned} -\sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} &= -\sum_{j\sim k} \bar{y}_j \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &\quad + \sum_{j\sim k} \bar{y}_k \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &= \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \end{aligned} \tag{S9.9}$$

Boundary terms on $\partial\Omega$ vanish because $\phi \in C_c^1(\Omega)^d$ has compact support in Ω.

Substep 2d: Identifying the Measure. Since the equation:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \tag{S9.10}$$

holds for all $\phi \in C_c^1(\Omega)^d$, the uniqueness part of the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) identifies $D\hat{f}_n$ uniquely as the measure:

$$D\hat{f}_n = \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k} \tag{S9.11}$$

where $\mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k}$ denotes the $(d-1)$-dimensional Hausdorff measure restricted to the shared boundary between leaves j and k. This is the formula stated in the lemma. □

S10. Proof of Lemma 6 (Derivative Consistency, BV).

S10.1. *What Needs to Be Shown.* That well-behavedness in the BV sense implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$, and the strict convergence upgrade under the additional condition $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$.

S10.2. *Proof Strategy.* The proof mirrors the subsequence argument of Lemma 4, replacing the Rellich–Kondrachov compact embedding with the BV compactness theorem and replacing strong L^p convergence with weak-* convergence of measures. We show that every subsequence of $\{D\hat{f}_n\}$ has a further subsequence converging weak-* to Df. By the metric space characterization of convergence in probability, this implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ in probability.

Suppose for contradiction that $D\hat{f}_n$ does not converge weak-* to Df in probability. Then there exist $\epsilon > 0$, $\delta > 0$, a test function $\phi \in C_0(\Omega)^d$ with $\|\phi\|_\infty \leq 1$, and a subsequence $\{n_k\}$ such that:

$$\mathbb{P}\left(\left| \int_\Omega \phi \cdot d(D\hat{f}_{n_k}) - \int_\Omega \phi \cdot d(Df) \right| > \epsilon \right) > \delta \quad \forall k \tag{S10.1}$$

We derive a contradiction by extracting a further subsequence along which weak-* convergence holds almost surely.

S10.3. *Part I: Weak-* Convergence.*

Step 1: Bound the Full BV Norm. To apply the BV compactness theorem we require a uniform bound on the full BV norm:

$$\left\|\hat{f}_n\right\|_{BV} = \left\|\hat{f}_n\right\|_{L^1} + |D\hat{f}_n|(\Omega) \tag{S10.2}$$

For the total variation term, the well-behaved condition gives:

$$\sup_n \mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] < \infty \tag{S10.3}$$

so $|D\hat{f}_n|(\Omega)$ is bounded in expectation and hence bounded almost surely along subsequences by Markov's inequality.

For the L^1 term, the consistency assumption $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$ implies that $\left\|\hat{f}_n\right\|_{L^1}$ is eventually bounded in probability. Since $f \in W^{1,p}(\Omega) \subset L^1(\Omega)$, we have $\|f\|_{L^1} < \infty$, and:

$$\left\|\hat{f}_n\right\|_{L^1} \leq \left\|\hat{f}_n - f\right\|_{L^1} + \|f\|_{L^1} \tag{S10.4}$$

which is bounded in probability. Combining the two terms:

$$\sup_n \left\|\hat{f}_n\right\|_{BV} < \infty \quad \text{almost surely} \tag{S10.5}$$

Step 2: Extract a Weakly- Convergent Subsequence.* Take any subsequence $\{\hat{f}_{n_k}\}$. By Step 1 it is bounded in $BV(\Omega)$ almost surely. The BV compactness theorem (Evans and Gariepy, 2015, Theorem 5.5) states that every sequence bounded in $BV(\Omega)$ for bounded Lipschitz Ω has a subsequence $\{\hat{f}_{n_{k_j}}\}$ and $g \in BV(\Omega)$ such that:

1. $\hat{f}_{n_{k_j}} \to g$ strongly in $L^1(\Omega)$
2. $D\hat{f}_{n_{k_j}} \xrightarrow{w*} Dg$ in $\mathcal{M}(\Omega)^d$

both holding almost surely. This is the BV analog of the Rellich–Kondrachov compact embedding: the BV compactness theorem gives simultaneously the L^1 convergence of the functions and the weak-* convergence of their gradient measures, in a single extraction.

Step 3: Identify the Limit. From item (1) of Step 2, $\hat{f}_{n_{k_j}} \to g$ strongly in $L^1(\Omega)$ almost surely. By the L^1 consistency assumption $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$, along the subsequence $\hat{f}_{n_{k_j}} \to f$ in L^1 in probability, hence almost surely along a further subsequence. By uniqueness of L^1 limits:

$$g = f \quad \mathcal{L}^d\text{-a.e. on } \Omega \tag{S10.6}$$

Therefore $Dg = Df$ as Radon measures, and from item (2) of Step 2:

$$D\hat{f}_{n_{k_j}} \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{almost surely} \tag{S10.7}$$

In particular, for the test function ϕ from the contradiction assumption:

$$\int_\Omega \phi \cdot d(D\hat{f}_{n_{k_j}}) \to \int_\Omega \phi \cdot d(Df) \quad \text{almost surely} \tag{S10.8}$$

Step 4: Derive the Contradiction. The almost sure convergence in Step 3 implies:

$$\mathbb{P}\left(\left|\int_\Omega \phi \cdot d(D\hat{f}_{n_{k_j}}) - \int_\Omega \phi \cdot d(Df)\right| > \epsilon\right) \to 0 \tag{S10.9}$$

This contradicts the assumption that the probability exceeds $\delta > 0$ for all k. Therefore the assumption was false.

Conclusion of Part I. Since every subsequence $\{D\hat{f}_{n_k}\}$ has a further subsequence converging weak-* to Df almost surely, the full sequence satisfies:

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{in probability} \tag{S10.10}$$

S10.4. *Part II: Strict Convergence Upgrade.* We now prove the additional claim: if $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$ then $D\hat{f}_n \to Df$ strictly in $\mathcal{M}(\Omega)^d$, meaning weak-* convergence holds and $|D\hat{f}_n|(\Omega) \to |Df|(\Omega)$.

Step 1: Lower Semicontinuity Gives One Direction. The total variation functional $\mu \mapsto |\mu|(\Omega)$ is lower semicontinuous with respect to weak-* convergence of Radon measures (Evans and Gariepy, 2015, Theorem 5.2). Since $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ by Part I:

$$|Df|(\Omega) \leq \liminf_{n\to\infty} |D\hat{f}_n|(\Omega) \tag{S10.11}$$

This gives the lower bound on the total variation of the limit.

Step 2: The Additional Condition Gives the Other Direction. By the additional assumption $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$, we have:

$$\limsup_{n\to\infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega) \tag{S10.12}$$

in expectation, hence in probability along subsequences.

Step 3: Pin the Total Variation. Step 1 gives the $\liminf$ inequality by lower semicontinuity of total variation under weak-* convergence, and Step 2 gives the $\limsup$ inequality from the additional assumption. Since $\liminf \leq \limsup$ always, combining gives:

$$|Df|(\Omega) \leq \liminf_{n\to\infty} |D\hat{f}_n|(\Omega) \leq \limsup_{n\to\infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega) \tag{S10.13}$$

All inequalities are therefore equalities and $|D\hat{f}_n|(\Omega) \to |Df|(\Omega)$ in probability. Combined with the weak-* convergence $D\hat{f}_n \xrightarrow{w*} Df$ established in Part I, this is precisely the definition of strict convergence in $\mathcal{M}(\Omega)^d$ (Ambrosio, Fusco and Pallara, 2000, Definition 3.13). □

REMARK 3. The parallel between Lemma 4 and Lemma 6 is exact at the structural level. In both cases: uniform boundedness in a strong space gives compactness via a compact embedding or compactness theorem; the consistency condition identifies the limit; and the subsequence argument promotes deterministic convergence to convergence in probability. The difference is entirely in the function spaces and the topology of convergence — strong L^p in the Sobolev case, weak-* of measures in the BV case — reflecting the fundamental difference in the inductive biases of smooth and piecewise constant estimators.

REMARK 4. The strict convergence upgrade requires the additional assumption that $\mathbb{E}[|D\hat{f}_n|(\Omega)]$ converge to $|Df|(\Omega)$. This is not implied by weak-* convergence alone. This is because weak-* convergence of measures does not in general imply convergence of total masses — mass can escape to the boundary or concentrate in ways invisible to continuous test functions. The lower semicontinuity of total variation (Evans and Gariepy, 2015, Theorem 5.2) gives the liminf inequality for free; pinning the limsup requires the additional assumption. In applications where only weak-* convergence is needed — as in the bootstrap corollaries — the additional assumption is not required.

S11. Proof of Theorem 3 (Pairing Identity).

PROOF. Write points as $x = (x_{-j}, x_j)$ and for fixed x_{-j} let $\Omega^{x_{-j}} = \{\tau \in \mathbb{R} : (x_{-j}, \tau) \in \Omega\}$ denote the one-dimensional section of the domain and $g^{x_{-j}}(\tau) = g(x_{-j}, \tau)$ the section of g. By the one-dimensional section theory of BV functions (Ambrosio, Fusco and Pallara, 2000, Section 3.11), for $\mathcal{H}^{d-1}$-almost every x_{-j} the section $g^{x_{-j}}$ of the precise representative is a good representative of a function of bounded variation on $\Omega^{x_{-j}}$, its distributional derivative $Dg^{x_{-j}}$ is the corresponding section of $D_j g$, and the total variations disintegrate:

$$\int_\Omega \phi \, dD_j g = \int \left(\int_{\Omega^{x_{-j}}} \phi(x_{-j}, \tau) \, dDg^{x_{-j}}(\tau) \right) d\mathcal{H}^{d-1}(x_{-j}) \tag{S11.1}$$

for every bounded Borel ϕ. For a good representative of a BV function of one variable, the fundamental theorem of calculus holds in the form $g^{x_{-j}}(b) - g^{x_{-j}}(a) = Dg^{x_{-j}}\big((a, b]\big)$ for all $a < b$ in $\Omega^{x_{-j}}$ that are not atoms of $Dg^{x_{-j}}$; since each section has at most countably many atoms, for Lebesgue-almost every x_j the pair $(x_j - h, x_j + h)$ avoids them. Hence for almost every $x \in \operatorname{supp}(w)$ (where the whole segment lies in Ω by the trimming condition):

$$g(x + he_j) - g(x - he_j) = Dg^{x_{-j}}\big((x_j - h,\, x_j + h]\big) \tag{S11.2}$$

Multiplying by $q(x)/2h$, integrating, and applying Tonelli on each section (justified since $|Dg^{x_{-j}}|$ is finite for a.e. section and q is bounded):

$$\begin{aligned}
&\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int dx_j \, \frac{q(x_{-j}, x_j)}{2h} \int \mathbf{1}\{\tau \in (x_j - h, x_j + h]\} \, dDg^{x_{-j}}(\tau) \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int dDg^{x_{-j}}(\tau) \, \frac{1}{2h} \int \mathbf{1}\{x_j \in [\tau - h, \tau + h)\} \, q(x_{-j}, x_j) \, dx_j \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int (K_h \star q)(x_{-j}, \tau) \, dDg^{x_{-j}}(\tau) \\
&= \int_\Omega (K_h \star q) \, dD_j g
\end{aligned} \tag{S11.3}$$

using $\tau \in (x_j - h, x_j + h] \Leftrightarrow x_j \in [\tau - h, \tau + h)$ in the second line, the definition of $K_h \star q$ (half-open versus closed windows agree up to Lebesgue-null sets) in the third, and (S11.1) in the last step. The piecewise constant case follows by substituting the explicit jump formula of Lemma 5 for $D_j g$.

For the corollary: if q extends to a uniformly continuous function of compact support on $\mathbb{R}^d$, then $K_h \star q \to q$ uniformly as $h \downarrow 0$, so:

$$\left| \int (K_h \star q) \, dD_j g - \int q \, dD_j g \right| \le \|K_h \star q - q\|_\infty \cdot |D_j g|(\Omega) \to 0 \tag{S11.4}$$

since $|D_j g|(\Omega) < \infty$. When $g \in W^{1,1}(\Omega)$, the measure $D_j g$ is absolutely continuous with density $\partial_j g$, so $\int q \, dD_j g = \int w \, \partial_j g \, p \, dx = \mathbb{E}[w \, \partial_j g(X)]$. □

S12. Proof of Theorem 4 (Neural Networks).

S12.1. *Setup and Notation.* Let $\hat{f}_n : \Omega \to \mathbb{R}$ be a feedforward neural network with L layers, weight matrices $W_\ell^{(n)} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, bias vectors $b_\ell^{(n)} \in \mathbb{R}^{d_\ell}$, and activation function $\sigma \in C^2(\mathbb{R})$. The network computes:

$$\hat{f}_n(x) = W_L^{(n)} z_{L-1} + b_L^{(n)} \tag{S12.1}$$

where the intermediate representations are defined recursively by:

$$z_\ell = \sigma\left(W_\ell^{(n)} z_{\ell-1} + b_\ell^{(n)}\right), \quad z_0 = x \tag{S12.2}$$

with σ applied elementwise. Denote the pre-activation at layer ℓ by $a_\ell = W_\ell^{(n)} z_{\ell-1} + b_\ell^{(n)}$ and the diagonal matrix of activation derivatives by $\Sigma'_\ell = \mathrm{diag}(\sigma'(a_\ell^{(1)}), \ldots, \sigma'(a_\ell^{(d_\ell)}))$ and similarly $\Sigma''_\ell = \mathrm{diag}(\sigma''(a_\ell^{(1)}), \ldots, \sigma''(a_\ell^{(d_\ell)}))$.

Define the product of weight matrices from layer k to layer ℓ as:

$$M_{\ell:k} = W_\ell^{(n)} \Sigma'_{\ell-1} W_{\ell-1}^{(n)} \Sigma'_{\ell-2} \cdots W_{k+1}^{(n)} \Sigma'_k \quad \text{for } \ell > k \tag{S12.3}$$

with the convention $M_{\ell:\ell} = \Sigma'_\ell$. This notation captures the Jacobian of the composition of layers from k to ℓ.

The proof proceeds in two steps. Step 1 establishes uniform $W^{2,p}(\Omega)$ boundedness. Step 2 applies Lemma 4 to conclude.

S12.2. *Step 1: Uniform $W^{2,p}$ Boundedness.* **Claim:** $\sup_n \left\|\hat{f}_n\right\|_{W^{2,p}(\Omega)} < \infty$.

Substep 1a: Bound on $\hat{f}_n$ in L^p. Since $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ and $f \in W^{1,p}(\Omega) \subset L^p(\Omega)$, the triangle inequality gives:

$$\left\|\hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n - f\right\|_{L^p} + \|f\|_{L^p} \tag{S12.4}$$

The first term converges to zero in probability and the second is finite since $f \in L^p(\Omega)$. Therefore $\left\|\hat{f}_n\right\|_{L^p}$ is eventually bounded in probability and this bound can be made uniform in n by passing to a subsequence or imposing a mild boundedness condition on the network output.

Substep 1b: First Derivative Bound. Differentiating $\hat{f}_n$ with respect to the input x via the chain rule through all L layers, the gradient at $x \in \Omega$ is:

$$\nabla \hat{f}_n(x) = \left(W_L^{(n)} M_{L-1:1}\right)^\top \tag{S12.5}$$

where $M_{L-1:1} = W_{L-1}^{(n)} \Sigma'_{L-2} W_{L-2}^{(n)} \cdots \Sigma'_1 W_1^{(n)}$ is the product of weight-activation Jacobians from layer 1 to layer $L-1$. More explicitly, for the j-th input coordinate:

$$\frac{\partial \hat{f}_n}{\partial x_j}(x) = \left(W_L^{(n)}\right)_{:} \cdot \prod_{\ell=L-1}^{1} \left(W_\ell^{(n)} \Sigma'_{\ell-1}\right) \cdot e_j \tag{S12.6}$$

where e_j is the j-th standard basis vector.

Taking the operator norm of this product and applying submultiplicativity:

$$\left|\frac{\partial \hat{f}_n}{\partial x_j}(x)\right| \leq \left\|W_L^{(n)}\right\| \cdot \prod_{\ell=1}^{L-1} \left\|W_\ell^{(n)}\right\| \cdot \left\|\Sigma'_\ell\right\| \leq \prod_{\ell=1}^{L} \left\|W_\ell^{(n)}\right\| \cdot \left\|\sigma'\right\|_\infty^{L-1} \tag{S12.7}$$

using that $\|\Sigma'_\ell\| = \max_i |\sigma'(a_\ell^{(i)})| \leq \|\sigma'\|_\infty < \infty$ since $\sigma \in C^2$. This bound holds pointwise for every $x \in \Omega$, so integrating over Ω:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\nabla \hat{f}_n\right\|_{L^\infty} \cdot |\Omega|^{1/p} \leq \prod_{\ell} \left\|W_\ell^{(n)}\right\| \cdot \left\|\sigma'\right\|_\infty^{L-1} \cdot |\Omega|^{1/p} \tag{S12.8}$$

Substep 1c: Second Derivative Bound. Differentiating the gradient formula once more with respect to x_i using the product rule and chain rule, the (i,j) entry of the Hessian $\nabla^2 \hat{f}_n(x)$ is:

(S12.9)
$$\frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x) = \sum_{\ell=1}^{L-1} \left(W_L^{(n)} M_{L-1:\ell+1} \right)_{:} \cdot \Sigma_\ell'' \cdot (M_{\ell:1} e_j) \cdot (M_{\ell:1} e_i)^\top \cdot \left(W_L^{(n)} M_{L-1:\ell+1} \right)_{:}^\top$$

where the sum runs over all layers ℓ at which the second derivative of σ is applied. The structure is: the forward pass from the input to layer ℓ contributes $M_{\ell:1} e_j$ for the j-th coordinate direction, the activation second derivative at layer ℓ contributes Σ_ℓ'', and the backward pass from layer ℓ to the output contributes $W_L^{(n)} M_{L-1:\ell+1}$.

Taking absolute values and applying submultiplicativity of operator norms:

$$\begin{aligned}
\left| \frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x) \right| &\leq \sum_{\ell=1}^{L-1} \left\| W_L^{(n)} M_{L-1:\ell+1} \right\|^2 \cdot \left\| \Sigma_\ell'' \right\| \cdot \| M_{\ell:1} \|^2 \\
&\leq \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L} \left\| W_k^{(n)} \right\| \right)^2 \cdot \| \sigma'' \|_\infty \cdot \left(\prod_{k=1}^{\ell} \left\| W_k^{(n)} \right\| \right)^2 \cdot \| \sigma' \|_\infty^{2(L-2)} \\
&\leq (L-1) \cdot \left(\prod_{\ell=1}^{L} \left\| W_\ell^{(n)} \right\| \right)^2 \cdot \| \sigma'' \|_\infty \cdot \| \sigma' \|_\infty^{2(L-2)} \qquad \text{(S12.10)}
\end{aligned}$$

using $\|\Sigma_\ell''\| \leq \|\sigma''\|_\infty < \infty$ since $\sigma \in C^2$, and collecting all weight norm factors into a single product. This bound holds pointwise for every $x \in \Omega$, so:

(S12.11)
$$\left\| \nabla^2 \hat{f}_n \right\|_{L^p} \leq \left\| \nabla^2 \hat{f}_n \right\|_{L^\infty} \cdot |\Omega|^{1/p} \leq (L-1) \cdot \left(\prod_{\ell} \left\| W_\ell^{(n)} \right\| \right)^2 \cdot \| \sigma'' \|_\infty \cdot \| \sigma' \|_\infty^{2(L-2)} \cdot |\Omega|^{1/p}$$

Combining Substeps 1a–1c. Summing the three bounds:

(S12.12)
$$\left\| \hat{f}_n \right\|_{W^{2,p}} = \left\| \hat{f}_n \right\|_{L^p} + \left\| \nabla \hat{f}_n \right\|_{L^p} + \left\| \nabla^2 \hat{f}_n \right\|_{L^p} \leq C \left(\prod_{\ell} \left\| W_\ell^{(n)} \right\|, \| \sigma' \|_\infty, \| \sigma'' \|_\infty, L, |\Omega| \right)$$

where C is a finite constant depending on the indicated quantities but not on n or x. By the assumption $\sup_n \prod_\ell \left\| W_\ell^{(n)} \right\| < \infty$ — which formalizes the effect of L^2 regularization during training (Bartlett, Foster and Telgarsky, 2017) — and the finiteness of $\|\sigma'\|_\infty$ and $\|\sigma''\|_\infty$ from $\sigma \in C^2$, the right-hand side is uniformly bounded in n. Therefore:

(S12.13)
$$\sup_n \left\| \hat{f}_n \right\|_{W^{2,p}} < \infty$$

This completes Step 1.

S12.3. *Step 2: L^p Consistency.* The L^p consistency condition $\left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0$ is the standard universal approximation result for smooth neural networks. Hornik, Stinchcombe and White (1990) establish that multilayer feedforward networks with smooth activation functions are dense in $W^{1,p}(\Omega)$ with respect to the $W^{1,p}$ norm, implying in particular density in $L^p(\Omega)$. Combined with standard empirical risk minimization results establishing that the training procedure converges to the best approximation in the function class, this gives $\left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0$.

S12.4. *Step 3: Applying Lemma 4.* By Step 1, $\sup_n \left\|\hat{f}_n\right\|_{W^{2,p}} < \infty$. By Step 2, $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$. These are precisely the two conditions of the well-behaved definition in the Sobolev sense (Definition 1). Lemma 4 then gives:

$$\left\|\nabla\hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{S12.14}$$

which is the conclusion of Theorem 4. □

REMARK 5. The weight norm assumption $\sup_n \prod_\ell \left\|W_\ell^{(n)}\right\| < \infty$ is the key regularity condition on the estimator. It formalizes the effect of L^2 weight decay regularization during training, which penalizes large weights and keeps the product of operator norms bounded. Under standard conditions on the regularization parameter and optimization procedure, this bound holds with high probability over the training randomness (Bartlett, Foster and Telgarsky, 2017). The condition is stated as an assumption of the theorem rather than derived from first principles because its verification depends on details of the training procedure that are outside the scope of the present analysis.

REMARK 6. Theorem 4 is stated for smooth activations $\sigma \in C^2$. For ReLU networks $\sigma \notin C^2$ and the Hessian calculation in Substep 1c does not apply directly. The first derivative exists almost everywhere with $\sigma' = \mathbf{1}_{(0,\infty)}$ bounded, so Substep 1b goes through unchanged. The distributional second derivative is $\sigma'' = \delta_0$, a Dirac mass at zero, placing $\hat{f}_n$ in a mixed $W^{1,p} \cap BV$ space rather than $W^{2,p}(\Omega)$. A modified version of Lemma 4 applicable in this mixed space, or a restriction to input distributions placing zero mass on the kink set $\{x : a_\ell(x) = 0\}$, would be required to handle this case. We leave this extension for future work.

S13. Proof of Theorem 5 (Gradient Boosting).

S13.1. *Setup and Notation.* By Proposition 1 of Bühlmann and Yu (2003), the L_2Boost estimator after m iterations can be represented as:

$$\hat{F}_m = (I - (I - S)^{m+1})Y \tag{S13.1}$$

where $S : \mathbb{R}^n \to \mathbb{R}^n$ is the base learner operator. The residual at step m is:

$$u_m = Y - \hat{F}_m = (I - S)^m Y \tag{S13.2}$$

so that each base learner fit is $\hat{f}_m = Su_m$ and the ensemble updates as $\hat{F}_m = \hat{F}_{m-1} + \hat{f}_m$.

By Proposition 2 of Bühlmann and Yu (2003), for a symmetric linear learner S with eigenvalues $\{\lambda_k; k = 1, \ldots, n\}$, the eigenvalues of the boosting operator $\mathcal{B}_m = I - (I - S)^{m+1}$ are $\{1 - (1 - \lambda_k)^{m+1}; k = 1, \ldots, n\}$.

Under the eigenvalue condition $0 < \lambda_k \leq 1$ for all k (Bühlmann and Yu, 2003, Theorem 1), let:

$$\rho = \max_k (1 - \lambda_k) \in [0, 1) \tag{S13.3}$$

Since $\lambda_k > 0$ for all k, we have $\rho < 1$. The bias decays exponentially fast: by Theorem 1(a) of Bühlmann and Yu (2003):

$$\|u_m\|_{L^2}^2 = \|(I - S)^m Y\|_{L^2}^2 \leq \rho^{2m} \|Y\|_{L^2}^2 \tag{S13.4}$$

In particular:

$$\|u_m\|_{L^2} \leq C\rho^m, \quad C = \|Y\|_{L^2} \tag{S13.5}$$

This geometric decay is the key engine for both cases of the proof.

The two cases of the theorem are handled separately.

S13.2. *Case (i): Smooth Base Learners.*

Step 1: $W^{2,p}$ Bound for a Single Base Learner. Each $\hat{f}_m = Su_m \in W^{2,p}(\Omega)$ by assumption on the base learner class. For shallow neural net base learners this follows from Section S12. For spline base learners it follows from standard spline approximation theory (Wahba, 1990). We require a uniform bound:

$$\sup_m \left\|\hat{f}_m\right\|_{W^{2,p}} \leq C' \|u_m\|_{L^2} < \infty \tag{S13.6}$$

where the bound on $\left\|\hat{f}_m\right\|_{W^{2,p}}$ in terms of $\|u_m\|_{L^2}$ follows from the boundedness of S as an operator from L^2 to $W^{2,p}$.

Step 2: Ensemble Bound via Geometric Decay. By linearity of the $W^{2,p}$ norm and the triangle inequality:

$$\left\|\hat{F}_m\right\|_{W^{2,p}} \leq \sum_{j=1}^{m} \left\|\hat{f}_j\right\|_{W^{2,p}} \leq C' \sum_{j=1}^{m} \|u_j\|_{L^2} \leq C'C \sum_{j=1}^{m} \rho^j \leq \frac{C'C\rho}{1-\rho} < \infty \tag{S13.7}$$

using the geometric decay (S13.5) and the fact that $\sum_{j=1}^{\infty} \rho^j = \rho/(1-\rho) < \infty$ for $\rho < 1$. The bound is uniform in m.

Step 3: Apply Lemma 4. L^p consistency is assumed. Uniform $W^{2,p}(\Omega)$ boundedness follows from Step 2. Lemma 4 gives $\left\|\nabla \hat{F}_m - \nabla f\right\|_{L^p} \xrightarrow{p} 0$. □

S13.3. *Case (ii): Tree Stump Base Learners.* Each base learner $\hat{f}_m = Su_m$ is a depth-1 tree (stump) of the form:

$$\hat{f}_m(x) = c_m^L \, \mathbf{1}[x_j \leq t_m] + c_m^R \, \mathbf{1}[x_j > t_m] \tag{S13.8}$$

for some split variable j, threshold t_m, and leaf constants:

$$c_m^L = \frac{1}{|I_L|} \sum_{i \in I_L} u_m(x_i), \quad c_m^R = \frac{1}{|I_R|} \sum_{i \in I_R} u_m(x_i) \tag{S13.9}$$

where $I_L = \{i : x_{ij} \leq t_m\}$ and $I_R = \{i : x_{ij} > t_m\}$ are the left and right leaf index sets.

Step 1: BV Norm of a Single Stump. By Lemma 5 the distributional gradient of $\hat{f}_m$ is:

$$D\hat{f}_m = (c_m^R - c_m^L) \cdot \nu_m \cdot \mathcal{H}^{d-1} \lfloor_{\{x_j = t_m\} \cap \Omega} \tag{S13.10}$$

so the total variation is:

$$|D\hat{f}_m|(\Omega) = |c_m^R - c_m^L| \cdot \mathcal{H}^{d-1}(\{x_j = t_m\} \cap \Omega) \tag{S13.11}$$

The geometric term satisfies $\mathcal{H}^{d-1}(\{x_j = t_m\} \cap \Omega) \leq \operatorname{diam}(\Omega)^{d-1}$, a constant depending only on Ω. For the response term, by Cauchy–Schwarz applied to the leaf means:

$$|c_m^R - c_m^L| \leq \frac{1}{|I_R|} \sum_{i \in I_R} |u_m(x_i)| + \frac{1}{|I_L|} \sum_{i \in I_L} |u_m(x_i)| \leq 2 \|u_m\|_{L^2} \tag{S13.12}$$

Therefore:

$$|D\hat{f}_m|(\Omega) \leq 2 \operatorname{diam}(\Omega)^{d-1} \cdot \|u_m\|_{L^2} \leq 2 \operatorname{diam}(\Omega)^{d-1} \cdot C\rho^m \tag{S13.13}$$

using the geometric decay (S13.5).

Step 2: Ensemble BV Bound via Geometric Series. By subadditivity of total variation:

(S13.14)
$$|D\hat{F}_m|(\Omega) \leq \sum_{j=1}^{m} |D\hat{f}_j|(\Omega) \leq 2\,\mathrm{diam}(\Omega)^{d-1} \cdot C \sum_{j=1}^{m} \rho^j \leq \frac{2\,\mathrm{diam}(\Omega)^{d-1} \cdot C\rho}{1-\rho} < \infty$$

The geometric series $\sum_{j=1}^{\infty} \rho^j = \rho/(1-\rho)$ converges because $\rho < 1$, which follows from the eigenvalue condition $\lambda_k > 0$ for all k. Taking expectations:

(S13.15)
$$\mathbb{E}[|D\hat{F}_m|(\Omega)] \leq \frac{2\,\mathrm{diam}(\Omega)^{d-1} \cdot \mathbb{E}[C]\rho}{1-\rho} < \infty$$

uniformly in m, where $\mathbb{E}[C] = \mathbb{E}[\|Y\|_{L^2}] < \infty$ by assumption.

Step 3: Apply Lemma 6. L^1 consistency is assumed. Uniform BV boundedness follows from Step 2. Lemma 6 gives $D\hat{F}_m \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. □

REMARK 7. The key step in both cases is the geometric decay $\|u_m\|_{L^2} \leq C\rho^m$, which follows from the eigenvalue condition $0 < \lambda_k \leq 1$ via Proposition 1 and Theorem 1 of Bühlmann and Yu (2003). This decay is much faster than any polynomial rate and implies that the geometric series $\sum_j \rho^j$ converges without any condition on step sizes. The eigenvalue condition is the operative regularity assumption for both cases of the theorem, replacing the step size conditions common in the boosting literature.

REMARK 8. When $\lambda_k \in \{0, 1\}$ for all k — as occurs for linear projection operators — Corollary 1 of Bühlmann and Yu (2003) shows that $\mathcal{B}_m \equiv S$ for all m, meaning boosting does not improve on the base learner. In this degenerate case $\rho = \max_k(1 - \lambda_k)$ takes the value 1 for any k with $\lambda_k = 0$, and the geometric series does not converge. The strict positivity condition $\lambda_k > 0$ excludes this case and corresponds to the requirement that the base learner has positive but incomplete explanatory power — the standard "weak learner" assumption in the boosting literature.

S14. Proof of Theorem 6 (Mondrian Forests).

S14.1. *Setup.* Let $\hat{f}_n = \frac{1}{B}\sum_{b=1}^{B} T_b$ where T_b are individual Mondrian trees. Each T_b is piecewise constant on a random partition $\mathcal{P}_b$ of $\Omega = [0, 1]^d$ generated by the Mondrian process $MP(\lambda_n, [0, 1]^d)$ independently of the data, with budget $\lambda_n \asymp n^{1/(d+2)}$ (Mourtada, Gaïffas and Scornet, 2017). The partition $\mathcal{P}_b$ consists of $K_{\lambda_n} + 1$ hyperrectangular cells, where K_{λ_n} is the number of splits (Roy and Teh, 2009).

S14.2. *Step 1: Each Tree is in BV.* Each T_b is piecewise constant on a finite partition and therefore in $BV(\Omega)$ by Lemma 5. The forest average $\hat{f}_n = \frac{1}{B}\sum_b T_b$ is a finite convex combination of BV functions, hence $\hat{f}_n \in BV(\Omega)$ with:

(S14.1)
$$|D\hat{f}_n|(\Omega) \leq \frac{1}{B}\sum_{b=1}^{B} |DT_b|(\Omega)$$

by convexity of the total variation functional (Ambrosio, Fusco and Pallara, 2000, Proposition 3.6).

S14.3. *Step 2: Uniform BV Boundedness via the Second Moment Argument.* **Claim:** $\sup_n \mathbb{E}[|D\hat{f}_n|(\Omega)] < \infty$.

The triangle inequality applied to the forest average is too crude because individual tree BV norms grow with n. Instead we work with the second moment and apply Cauchy–Schwarz:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] \le \sqrt{\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big]} \tag{S14.2}$$

It therefore suffices to show $\mathbb{E}[|D\hat{f}_n|(\Omega)^2] < \infty$ uniformly in n.

Substep 2a: Expand the Second Moment. Writing $D\hat{f}_n = \frac{1}{B}\sum_b DT_b$ and expanding:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big] = \frac{1}{B^2}\mathbb{E}\left[\left|\sum_b DT_b\right|(\Omega)^2\right] = \frac{1}{B^2}\sum_{b,b'}\mathbb{E}\left[\int \phi\cdot d(DT_b)\cdot\int \phi\cdot d(DT_{b'})\right] \tag{S14.3}$$

for a suitable test function ϕ with $\|\phi\|_\infty \le 1$. The double sum splits into diagonal terms $b = b'$ and cross terms $b \ne b'$.

Substep 2b: Cross Terms Vanish. For $b \ne b'$ the partitions $\mathcal{P}_b$ and $\mathcal{P}_{b'}$ are independent of each other and of the data. By independence:

$$\mathbb{E}\left[\int \phi\cdot d(DT_b)\cdot\int \phi\cdot d(DT_{b'})\right] = \mathbb{E}\left[\int \phi\cdot d(DT_b)\right]\cdot\mathbb{E}\left[\int \phi\cdot d(DT_{b'})\right] \tag{S14.4}$$

We claim each factor is zero. By Lemma 5 and the explicit formula for DT_b:

$$\mathbb{E}\left[\int \phi\cdot d(DT_b)\right] = \mathbb{E}\left[\sum_{j\sim k}(\bar{y}_k^b - \bar{y}_j^b)\int_{\partial R_j^b\cap\partial R_k^b}\phi\cdot\nu_{jk}\,d\mathcal{H}^{d-1}\right] \tag{S14.5}$$

Since $\mathcal{P}_b$ is independent of the data, we condition on the partition and take the expectation over the data first:

$$= \mathbb{E}_{\mathcal{P}_b}\left[\sum_{j\sim k}\mathbb{E}_{\text{data}}\Big[\bar{y}_k^b - \bar{y}_j^b\Big]\int_{\partial R_j^b\cap\partial R_k^b}\phi\cdot\nu_{jk}\,d\mathcal{H}^{d-1}\right] \tag{S14.6}$$

The leaf means satisfy $\mathbb{E}_{\text{data}}[\bar{y}_j^b] \to f(\bar{x}_j^b)$ as $n\to\infty$, where $\bar{x}_j^b$ is the center of leaf R_j^b. For adjacent leaves j and k, by the Lipschitz continuity of f:

$$\mathbb{E}_{\text{data}}\Big[\bar{y}_k^b - \bar{y}_j^b\Big] \approx f(\bar{x}_k^b) - f(\bar{x}_j^b) = O(D_{\lambda_n}) \tag{S14.7}$$

where D_{λ_n} is the cell diameter. As $\lambda_n\to\infty$ the cell diameter $D_{\lambda_n}\to 0$ and each term vanishes. More precisely, since $f\in W^{1,p}(\Omega)$ with $p > d$, the Sobolev embedding $W^{1,p}\hookrightarrow C^{0,\alpha}$ for $\alpha = 1 - d/p$ gives:

$$\left|\mathbb{E}_{\text{data}}\Big[\bar{y}_k^b - \bar{y}_j^b\Big]\right| \le C\cdot\operatorname{diam}(R_j^b\cup R_k^b)^\alpha \le C\cdot D_{\lambda_n}^\alpha \to 0 \tag{S14.8}$$

Therefore $\mathbb{E}[\int\phi\cdot d(DT_b)]\to 0$ and the cross terms vanish asymptotically. For n large enough the cross terms are negligible and the second moment is dominated by the diagonal terms.

Substep 2c: Bound the Diagonal Terms. For the diagonal terms $b = b'$:

$$\mathbb{E}\big[|DT_b|(\Omega)^2\big] = \mathbb{E}\left[\left(\sum_{j\sim k} |\bar{y}_k^b - \bar{y}_j^b| \cdot \mathcal{H}^{d-1}(\partial R_j^b \cap \partial R_k^b)\right)^2\right] \tag{S14.9}$$

Expanding the square and separating geometric and response terms using the independence of $\mathcal{P}_b$ from the data:

$$\mathbb{E}\big[|DT_b|(\Omega)^2\big] \le \mathbb{E}_{\mathcal{P}_b}\big[K_{\lambda_n}^2\big] \cdot \sup_{j\sim k} \mathbb{E}_{\text{data}}\Big[|\bar{y}_k^b - \bar{y}_j^b|^2\Big] \tag{S14.10}$$

where K_{λ_n} is the number of adjacent pairs, which equals the number of splits.

Geometric Factor. The Mondrian process generates cuts according to a recursive tree structure. By the conditional independence of subtrees (Roy and Teh, 2009), the number of splits satisfies the distributional recursion:

$$K_{\lambda_n} = 1 + K_{\lambda_n, L} + K_{\lambda_n, R} \tag{S14.11}$$

where $K_{\lambda_n,L}$ and $K_{\lambda_n,R}$ are the split counts in the left and right subtrees, conditionally independent given the first cut. From Mourtada, Gaïffas and Scornet (2017):

$$\mathbb{E}[K_{\lambda_n}] \le (e(\lambda_n + 1))^d = O(n^{d/(d+2)}) \tag{S14.12}$$

Taking the second moment via the recursion and using conditional independence:

$$\mathbb{E}[K_{\lambda_n}^2] = \mathbb{E}[K_{\lambda_n}] + \mathbb{E}[K_{\lambda_n}]^2 = O(n^{d/(d+2)}) + O(n^{2d/(d+2)}) = O(n^{2d/(d+2)}) \tag{S14.13}$$

where the first equality uses the Poisson-like variance structure of the Mondrian process implied by the conditional independence recursion.

Response Factor. For adjacent leaves j and k, by Cauchy–Schwarz on the sample means:

$$\mathbb{E}_{\text{data}}\Big[|\bar{y}_k^b - \bar{y}_j^b|^2\Big] \le 4\,\mathbb{E}_{\text{data}}\Big[\|r^{(b)}\|_{L^2}^2\Big] \tag{S14.14}$$

where $r^{(b)}$ denotes the residual within the relevant cells. Using the cell diameter bound of Roy and Teh (2009):

$$\mathbb{E}[D_{\lambda_n}(x)^2] \le \frac{4d}{\lambda_n^2} = O(n^{-2/(d+2)}) \tag{S14.15}$$

and the Hölder continuity of $f \in W^{1,p}$ with $p > d$ giving $\alpha = 1 - d/p$:

$$\mathbb{E}_{\text{data}}\Big[|\bar{y}_k^b - \bar{y}_j^b|^2\Big] \le C^2 \left(\mathbb{E}[D_{\lambda_n}(x)^2]\right)^\alpha \le C^2 \left(\frac{4d}{\lambda_n^2}\right)^\alpha = O(n^{-2\alpha/(d+2)}) \tag{S14.16}$$

Combining the Two Factors.

$$\mathbb{E}\big[|DT_b|(\Omega)^2\big] \le O(n^{2d/(d+2)}) \cdot O(n^{-2\alpha/(d+2)}) = O(n^{2(d-\alpha)/(d+2)}) \tag{S14.17}$$

Therefore:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big] = \frac{1}{B^2}\sum_b \mathbb{E}\big[|DT_b|(\Omega)^2\big] = \frac{1}{B} O(n^{2(d-\alpha)/(d+2)}) \tag{S14.18}$$

For this to be bounded uniformly in n we require:

$$B \gg n^{2(d-\alpha)/(d+2)} \tag{S14.19}$$

This is a condition on the number of trees relative to the dimension and smoothness of f. Under this condition:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] \le \sqrt{\mathbb{E}\Big[|D\hat{f}_n|(\Omega)^2\Big]} = \sqrt{\frac{1}{B} O(n^{2(d-\alpha)/(d+2)})} < \infty \tag{S14.20}$$

uniformly in n, completing the uniform BV boundedness claim.

REMARK 9. The condition $B \gg n^{2(d-\alpha)/(d+2)}$ requires the number of trees to grow with n at a rate depending on dimension d and Hölder exponent $\alpha = 1 - d/p$. For α close to 1 — achieved when $p \gg d$, meaning f is highly smooth — the condition is mild and a slowly growing number of trees suffices. For α small — meaning f has limited smoothness or d is large — more trees are required. In practice random forests with $B = 500$ or $B = 1000$ trees satisfy this condition for moderate d and n.

S14.4. *Step 3: Apply Lemma 6.* The L^1 consistency condition follows from the L^2 consistency result of Mourtada, Gaïffas and Scornet (2017) under the budget $\lambda_n \asymp n^{1/(d+2)}$ by the Cauchy–Schwarz inequality on the bounded domain Ω:

$$\left\|\hat{f}_n - f\right\|_{L^1} \le |\Omega|^{1/2} \left\|\hat{f}_n - f\right\|_{L^2} \xrightarrow{p} 0 \tag{S14.21}$$

Uniform BV boundedness follows from Step 2 under the condition $B \gg n^{2(d-\alpha)/(d+2)}$. Lemma 6 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. □

REMARK 10. Theorem 6 is stated for Mondrian forests where the partition is generated independently of the data. The independence is used critically in the decomposition of Substep 2a and in the cross term cancellation of Substep 2b. For Breiman forests with data-dependent splits the partition geometry and response variation are coupled — splits are chosen to maximize response differences across boundaries, which is precisely the worst case for the BV bound, and the cross term cancellation argument fails because the jump directions are not independent across trees. Extending the result to Breiman forests requires additional control over this coupling and is left as future work.

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